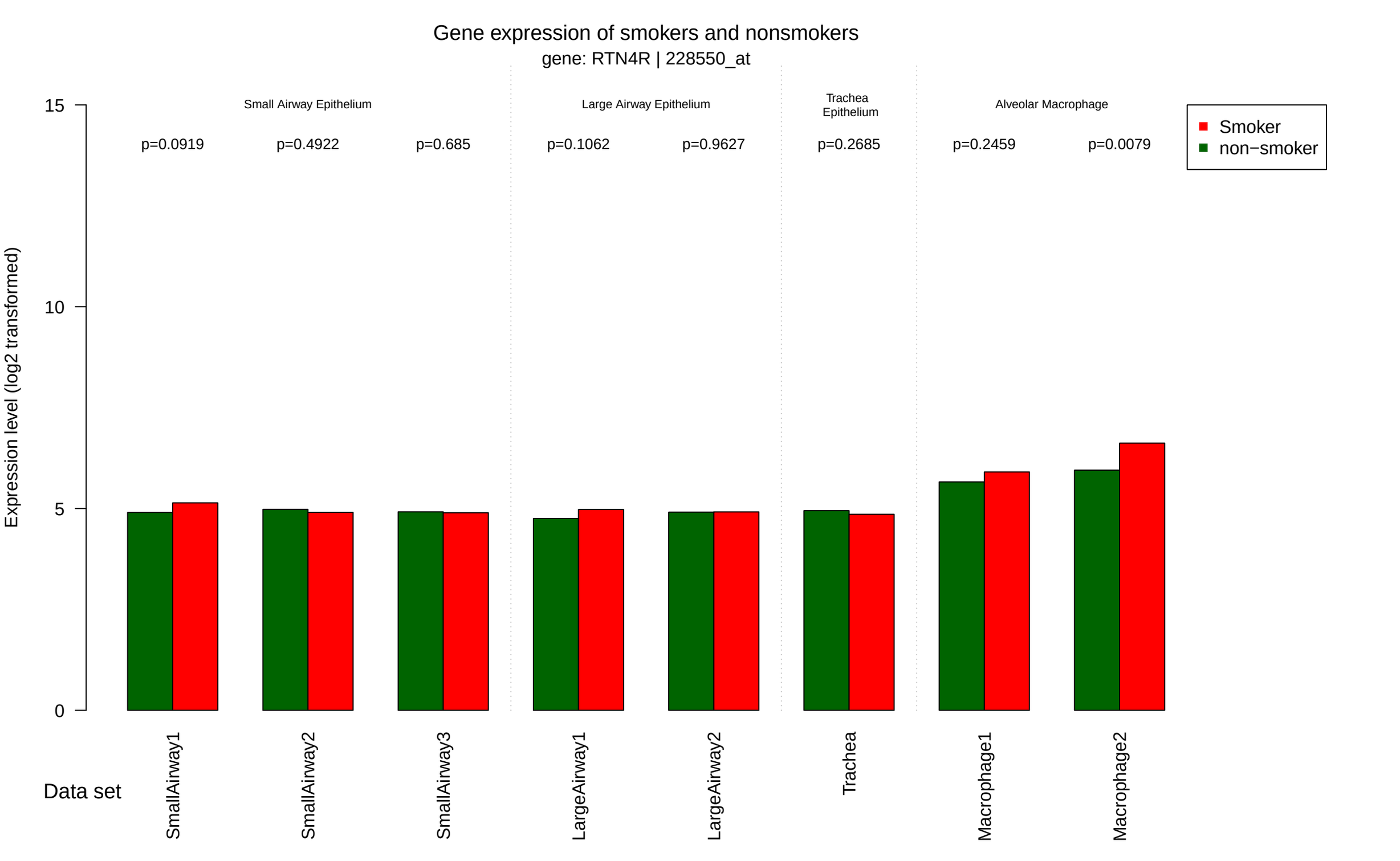
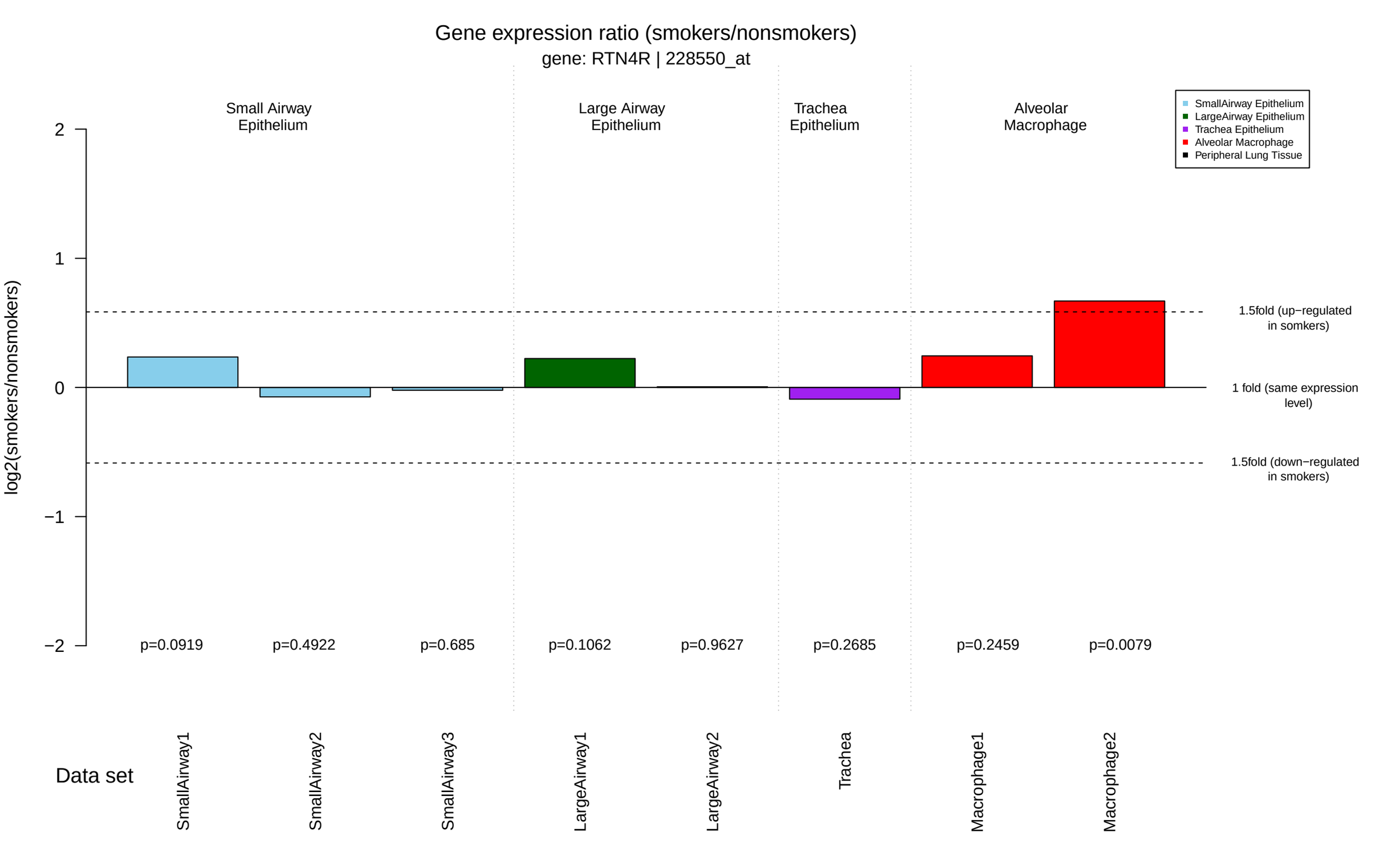
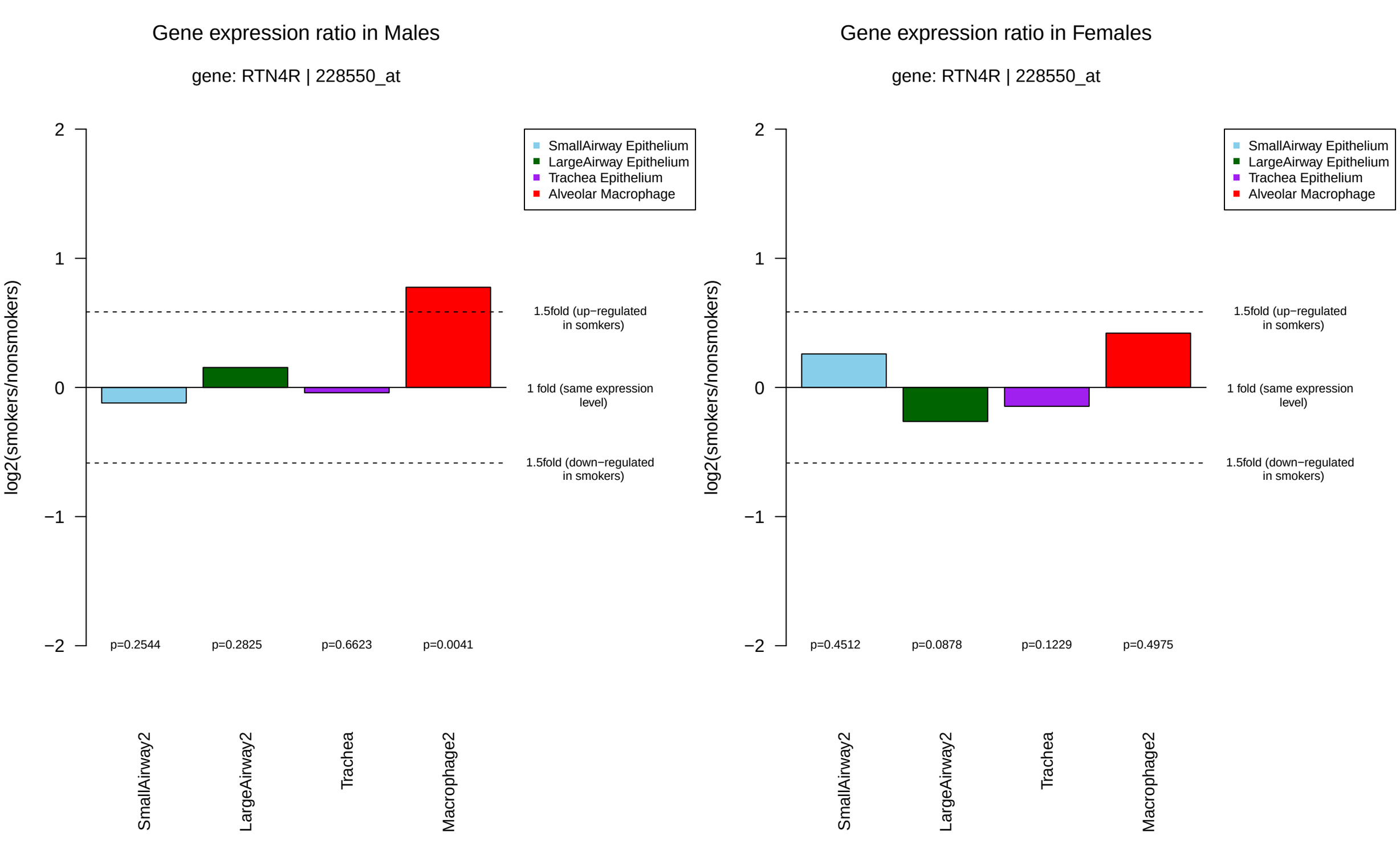
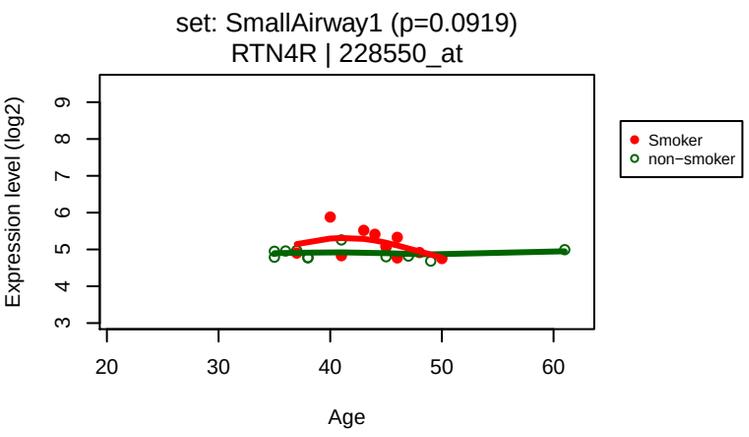
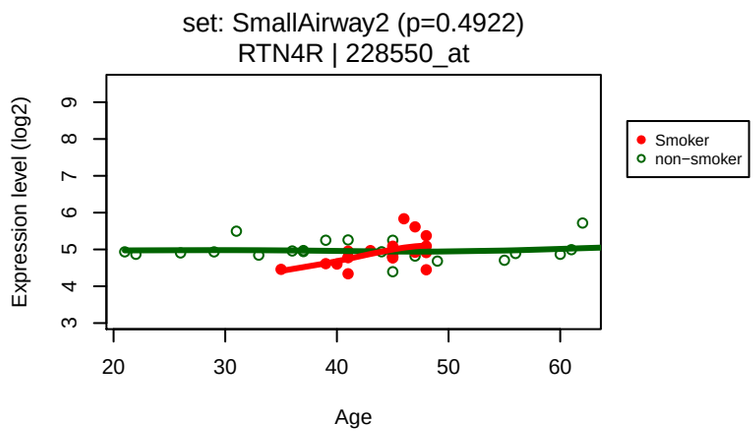
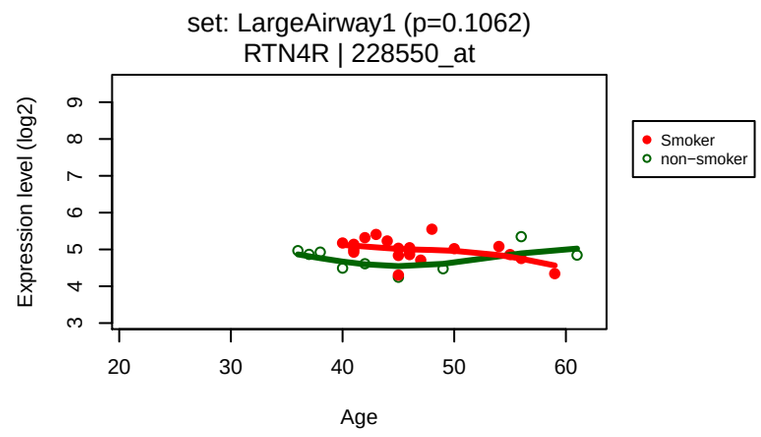

Gene expression of smokers and nonsmokers
gene: RTN4R | 228550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0919
p=0.4922
p=0.685
p=0.1062
p=0.9627
p=0.2685
p=0.2459
p=0.0079
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

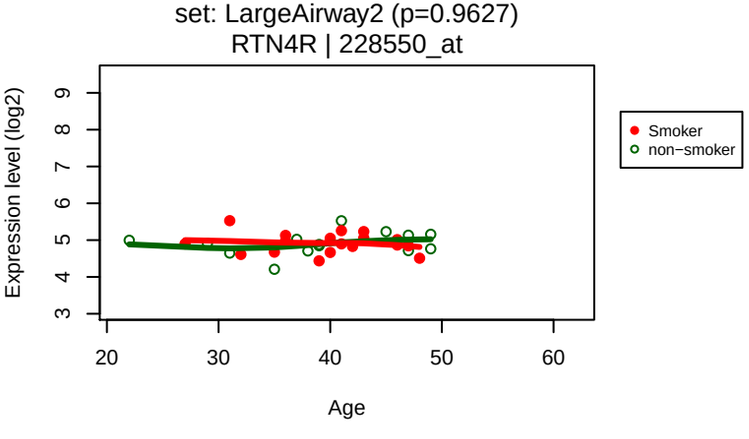

Gene expression ratio (smokers/nonsmokers)
gene: RTN4R | 228550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0919
p=0.4922
p=0.685
p=0.1062
p=0.9627
p=0.2685
p=0.2459
p=0.0079
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

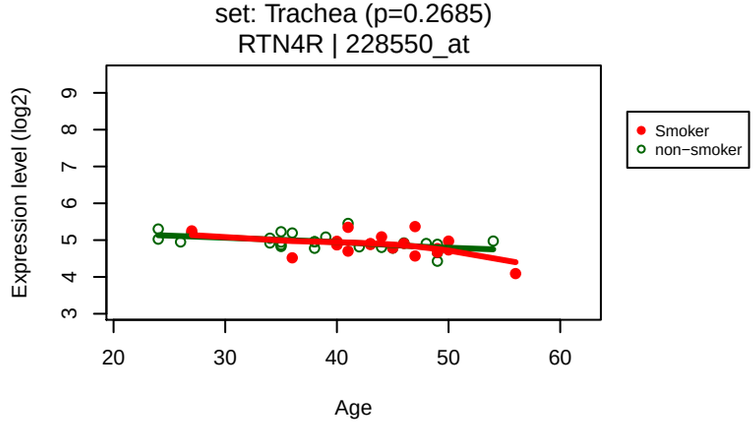

Gene expression ratio in Males
gene: RTN4R | 228550_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2544
p=0.2825
p=0.6623
p=0.0041
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RTN4R | 228550_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4512
p=0.0878
p=0.1229
p=0.4975
SmallAirway2
LargeAirway2
Trachea
Macrophage2

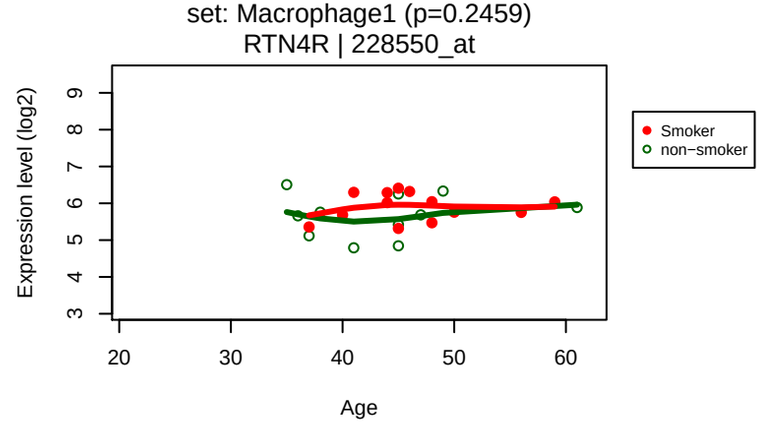

set: SmallAirway1 (p=0.0919)
RTN4R | 228550_at
Expression level (log2)
Age
Smoker
non−smoker

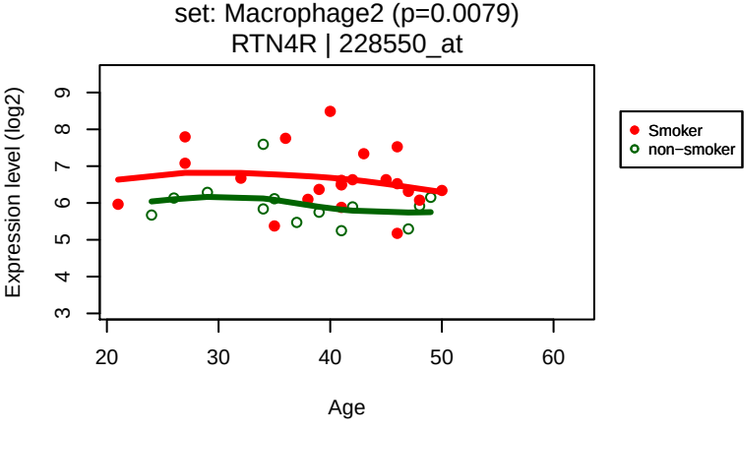

set: SmallAirway2 (p=0.4922)
RTN4R | 228550_at
Expression level (log2)
Age
Smoker
non−smoker

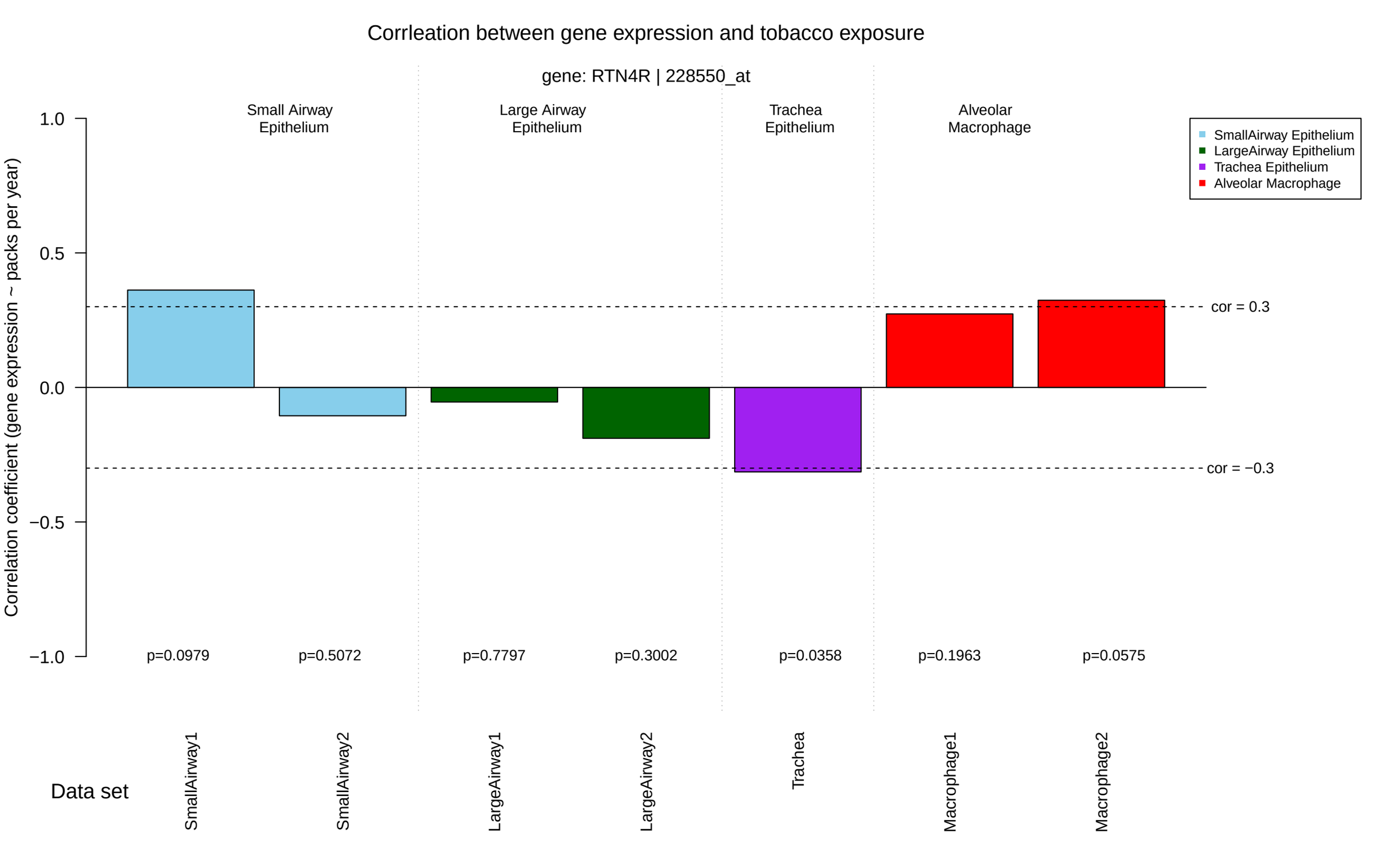

set: LargeAirway1 (p=0.1062)
RTN4R | 228550_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9627)
RTN4R | 228550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2685)
RTN4R | 228550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2459)
RTN4R | 228550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0079)
RTN4R | 228550_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RTN4R | 228550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0979
p=0.5072
p=0.7797
p=0.3002
p=0.0358
p=0.1963
p=0.0575
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2